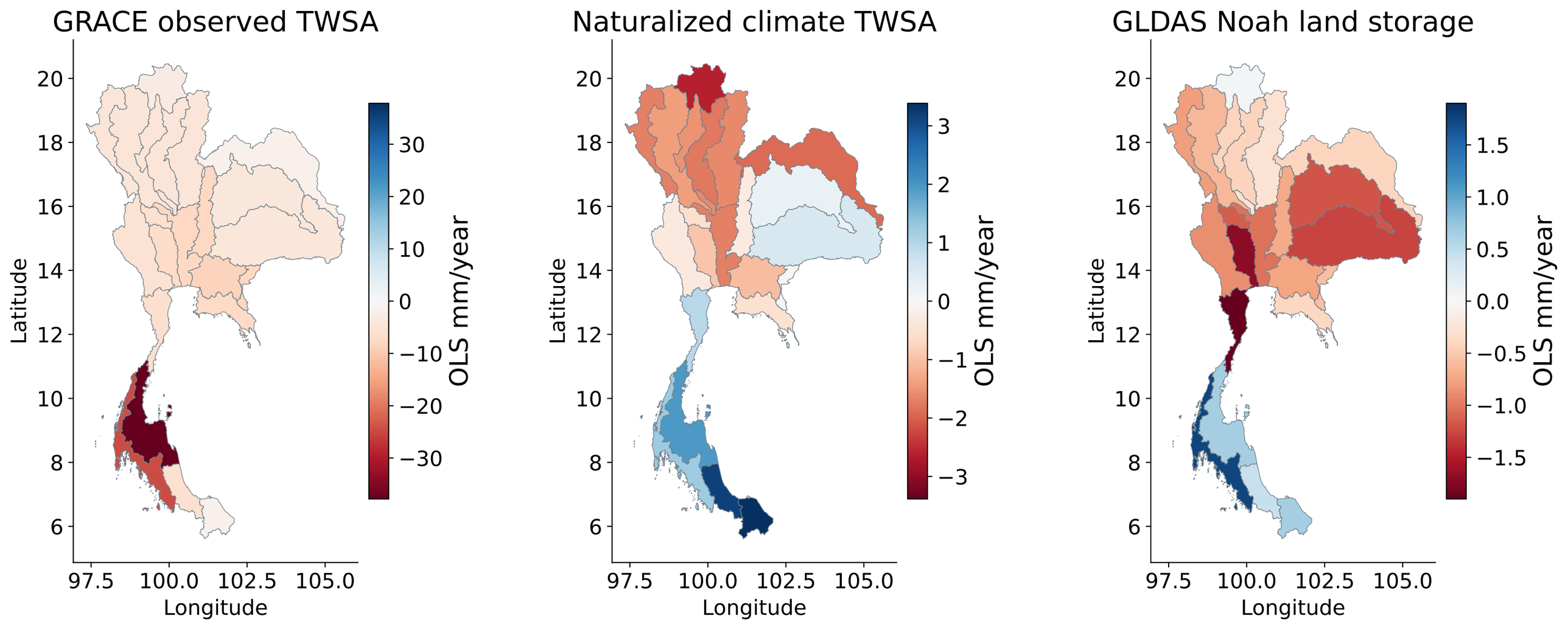
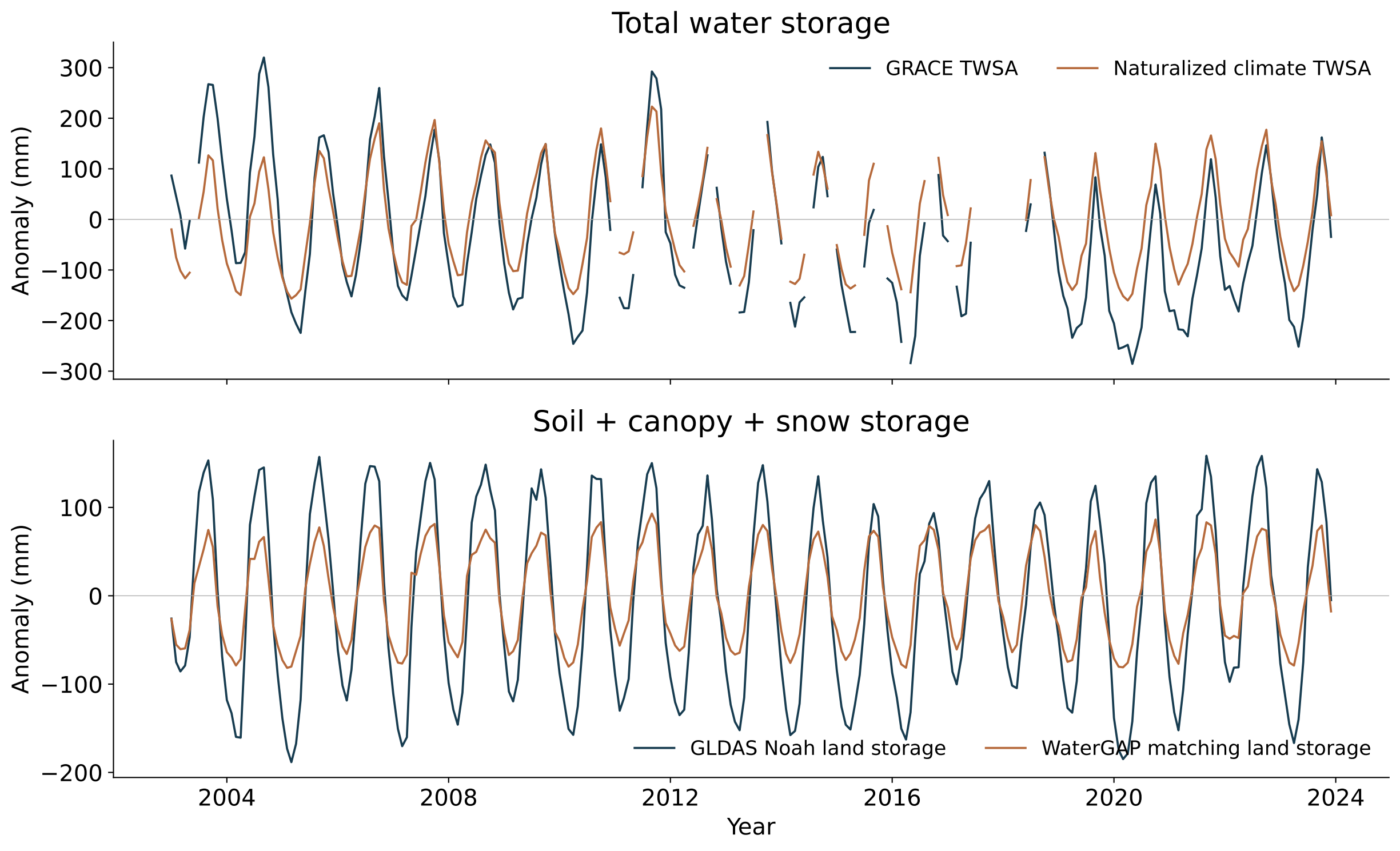

GRACE observed TWSA
Naturalized climate TWSA
GLDAS Noah land storage
Latitude
Longitude
OLS mm/year
20
18
16
14
12
10
8
6
97.5
100.0
102.5
105.0
30
20
10
0
−10
−20
−30
3
2
1
0
−1
−2
−3
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5


Total water storage
GRACE TWSA
Naturalized climate TWSA
Anomaly (mm)
300
200
100
0
−100
−200
−300
Soil + canopy + snow storage
GLDAS Noah land storage
WaterGAP matching land storage
Anomaly (mm)
100
0
−100
−200
2004
2008
2012
2016
2020
2024
Year

Storage anomaly (mm)

01 Salawin
02 North Khong
03 Northeast Khong
04 Chi
05 Mun
06 Ping
07 Wang
08 Yom
09 Nan
10 Chao Phraya
11 Sakae Krang
12 Pasak
13 Tha Chin
14 Mae Klong
15 Bang Pakong
16 Tonle Sap
17 East Coast Gulf
18 Phetchaburi - Pr
19 Peninsula - Uppe
20 Thale Sap Songkh
21 Peninsula - Lowe
22 Peninsula - West

Naturalized parent
Climate emulator

Year · full basin names are retained in the supplied CSV tables

01 Salawin

02 North Khong

03 Northeast Khong

04 Chi

05 Mun

06 Ping

07 Wang

08 Yom

09 Nan

10 Chao Phraya

11 Sakae Krang

12 Pasak

13 Tha Chin

14 Mae Klong

15 Bang Pakong

16 Tonle Sap

17 East Coast Gulf

18 Phetchaburi - Pr

19 Peninsula - Uppe

20 Thale Sap Songkh

21 Peninsula - Lowe

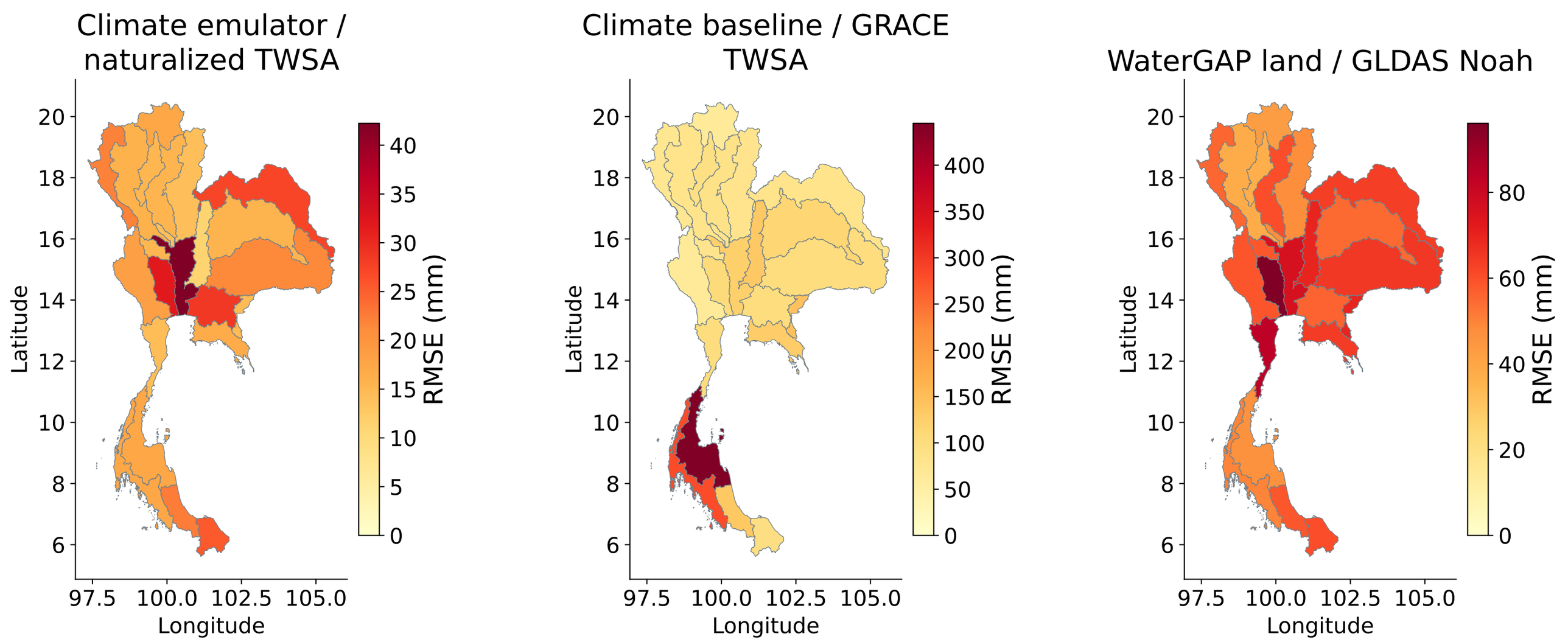
22 Peninsula - West

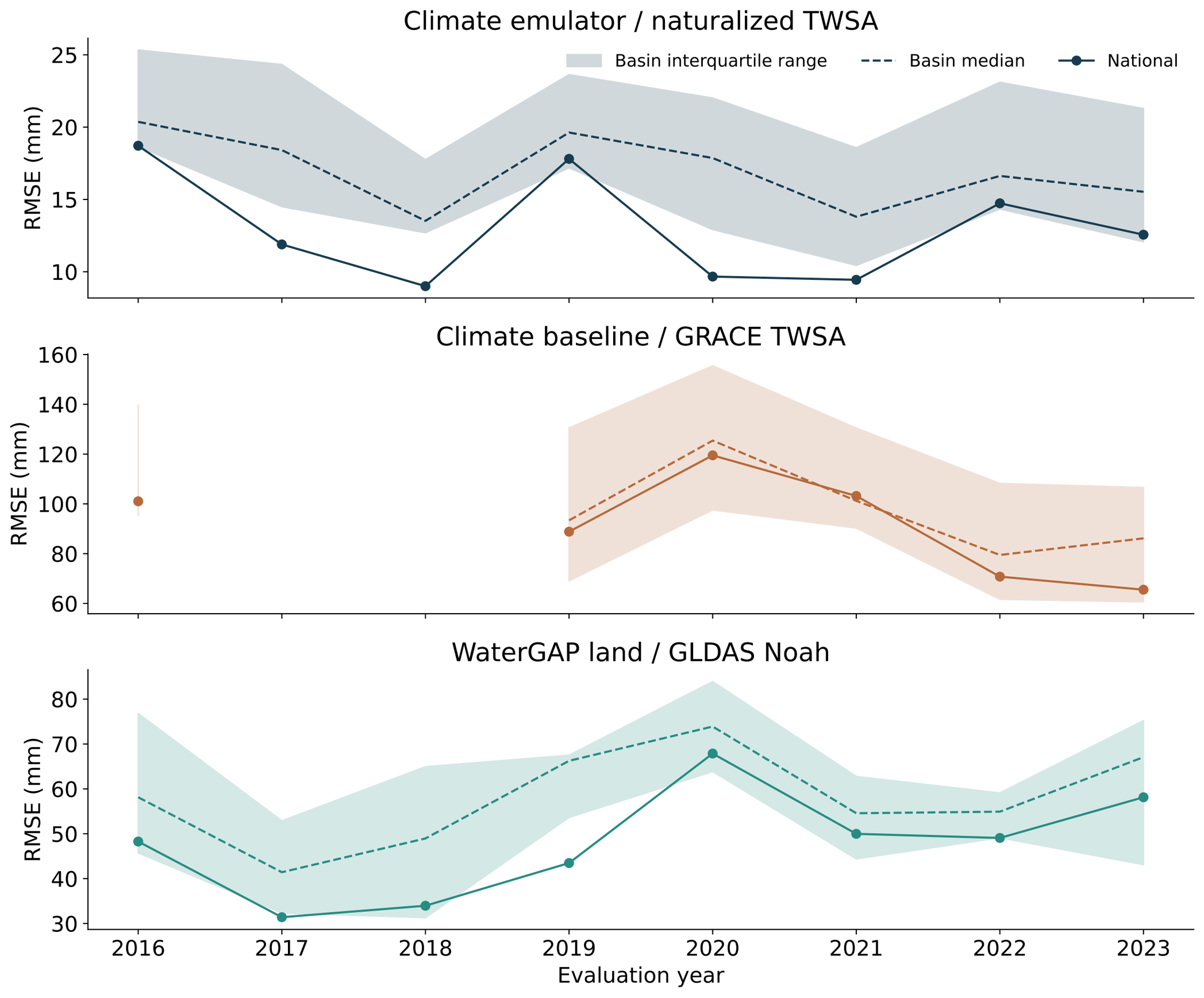
Storage anomaly (mm)

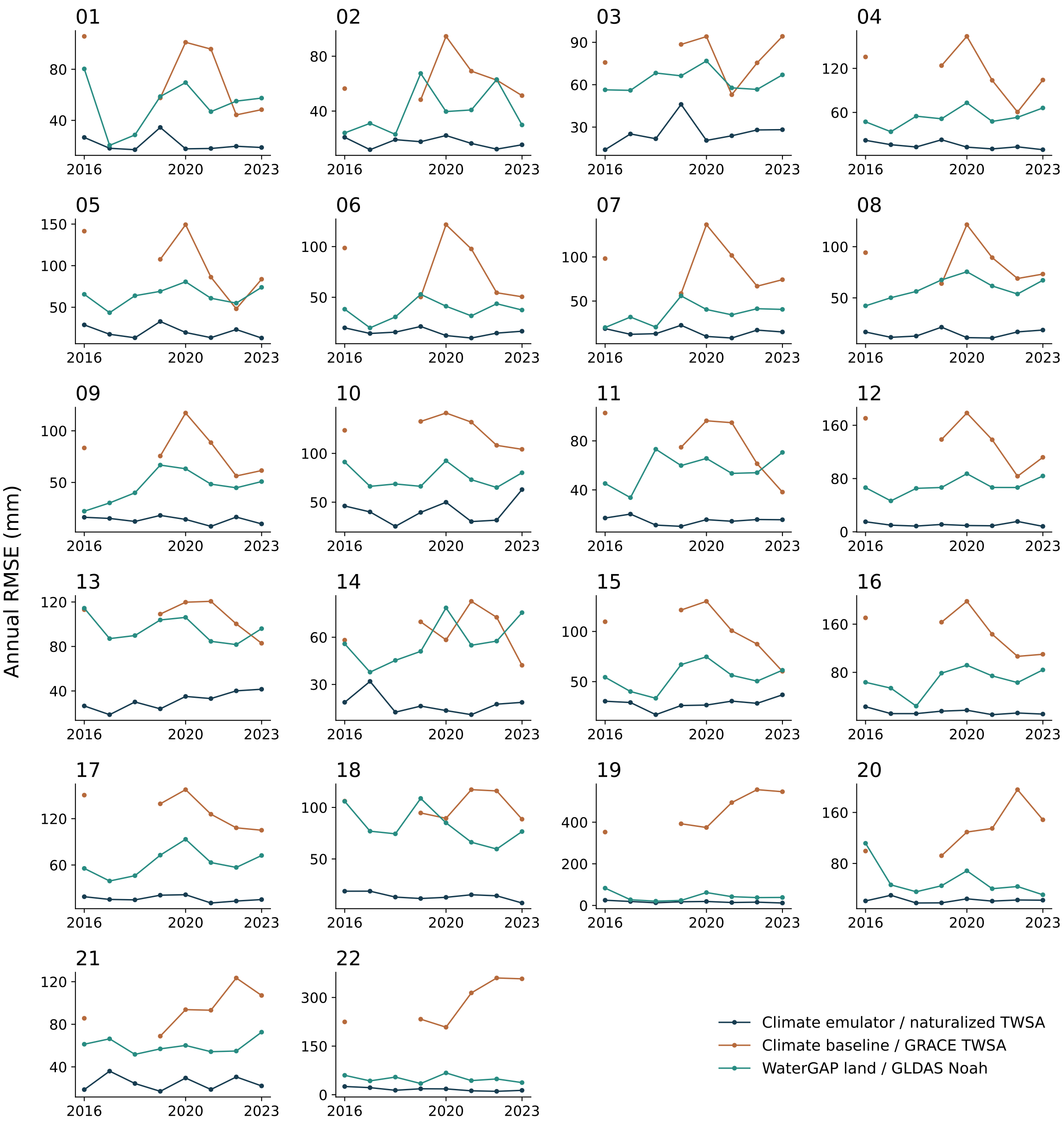
GLDAS land storage

WaterGAP matching land storage

Year · full basin names are retained in the supplied CSV tables


Climate emulator / naturalized TWSA
Climate baseline / GRACE TWSA
WaterGAP land / GLDAS Noah
Latitude
Longitude
RMSE (mm)
20
18
16
14
12
10
8
6
97.5
100.0
102.5
105.0
0
5
10
15
20
25
30
35
40
50
100
150
200
250
300
350
400
20
40
60
80


Climate emulator / naturalized TWSA
Basin interquartile range
Basin median
National
RMSE (mm)
25
20
15
10
Climate baseline / GRACE TWSA
RMSE (mm)
160
140
120
100
80
60
WaterGAP land / GLDAS Noah
RMSE (mm)
80
70
60
50
40
30
2016
2017
2018
2019
2020
2021
2022
2023
Evaluation year


01
02
03
04
05
06
07
08
09
10
11
12
13
14
15
16
17
18
19
20
21
22
Annual RMSE (mm)
2016
2020
2023
Climate emulator / naturalized TWSA
Climate baseline / GRACE TWSA
WaterGAP land / GLDAS Noah
Year · basin codes match the supplied CSV names